Badreddine Benfettouma

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Research interests

Digital integrated-circuit design and RISC-V microarchitecture; formal verification of hardware; energy-efficient hardware for machine-learning inference; open-source EDA flows and open PDKs.

Education

Licence 3 (third year of the Bachelor), Electronics 2026–2027
Université Bourgogne Europe, Dijon, France

Licence 1 and 2, Computer Science and Electronics track 2024–2026
Université de Bourgogne (Université Bourgogne Europe since January 2025), Dijon, France
Selected grades (out of 20): computer architecture 14.2 (lab 18.1), signal processing 17.5, scientific computing 20.

The French *Licence* is a three-year bachelor's degree (180 ECTS).

Projects

RV32I-Alpha: a two-stage RISC-V processor core 2026
VHDL, SystemVerilog; source code

- Two-stage pipelined core implementing the 37 instructions of the RV32I base ISA, with traps on illegal instructions and misaligned accesses and a data-memory port that accepts wait states. Written in VHDL, then ported to SystemVerilog.
- Formal verification: riscv-formal instruction and consistency checks (bounded model checking, 10–15 cycles) and custom SymbiYosys properties for the memory ports; mutation testing (8 injected bugs, all detected); Yosys proof that the VHDL and SystemVerilog versions are sequentially equivalent; the official riscv-tests `rv32ui` suite passes.
- FPGA: built for the Digilent Basys 3 (Artix-7, not yet tested on the board); the core alone uses 675 LUTs and meets 100 MHz (Vivado).
- ASIC: hardened to a DRC- and LVS-clean GDSII macro in SkyWater SKY130 with LibreLane (power grid, clock tree, sign-off; 0.22 mm^2, 40 MHz at the typical corner).

Area and energy accounting of logic-gate neural networks 2026
Independent project; Python, PyTorch, Yosys, OpenROAD, OpenSTA; source code

- Open flow: trained network, SKY130 netlist and placement, gate-level simulation on real images, power from the recorded switching activity; the netlist matches the software model bit for bit.
- Counted what comparisons usually leave out (input register, vote-counting head, wires): 78 099 $\mu\mathrm{m}^2$, 3.1 nJ per inference, 84 % of the power in flip-flops.

4-bit weight formats for LLM inference hardware 2026
Independent project; Python, PyTorch, Hugging Face Transformers; source code

- Perplexity of five 4-bit weight formats on three LLMs. At equal quality, trellis coding saves 11–17 % of the bits against NVFP4 but only 6–9 % against a rotated Lloyd-Max format.

Technical skills

HDLs	VHDL, SystemVerilog, Verilog
Verification	SymbiYosys, riscv-formal, Yosys equivalence checking, GHDL, Icarus Verilog
Implementation	Vivado (Artix-7); LibreLane, Yosys, OpenROAD, OpenSTA, Magic, KLayout (SKY130)
Circuits	ngspice with SKY130 device models, Monte-Carlo mismatch analysis
Programming	Python (NumPy, PyTorch, Hugging Face Transformers), Tcl, RISC-V assembly; Linux, Git

Languages

French (native) · English (C1) · Japanese (learning) · Algerian Arabic (spoken)